

PH101: 2023

Quantum Mechanics

Lecture 1

Subhadya Bhattacharya

Syllabus and References

Another significant departure from the known classical physics

Quantum Mechanics: Two-slit experiment. De Broglie's hypothesis. Uncertainty Principle, wave function and wave packets, phase and group velocities. Schrödinger Equation. Probabilities and Normalization. Expectation values. Eigenvalues and eigenfunctions.

Applications in one dimension: Particle in a box, Finite Potential well, Harmonic oscillator.

Text / References:

1. R. Eisberg and R. Resnick, *Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles*, 2nd Ed., John-Wiley, 1985.
2. E. W. Wichman, *Quantum Physics*, Berkeley Physics Course Vol. 4, Tata McGraw Hill (Indian Edition), 2008
3. D. J. Griffiths, *Introduction to Quantum Mechanics*, Pearson, 2005.
4. R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics, Vol. III*, Pearson, 2006.
5. A. Beiser, *Concepts of Modern Physics*, Tata McGraw-Hill, New Delhi, 1995.
6. S. Gasiorowicz, *Quantum Physics*, John Wiley (Asia), 2000.

We will essentially cover the topic with seven lectures and three tutorials

Classical Physics: Key points

In Newtonian Physics, the laws are written in terms of particle trajectories.

A particle is an indivisible object, whose variety of properties can be measured, including position and momentum. The same is true for a system of particles.

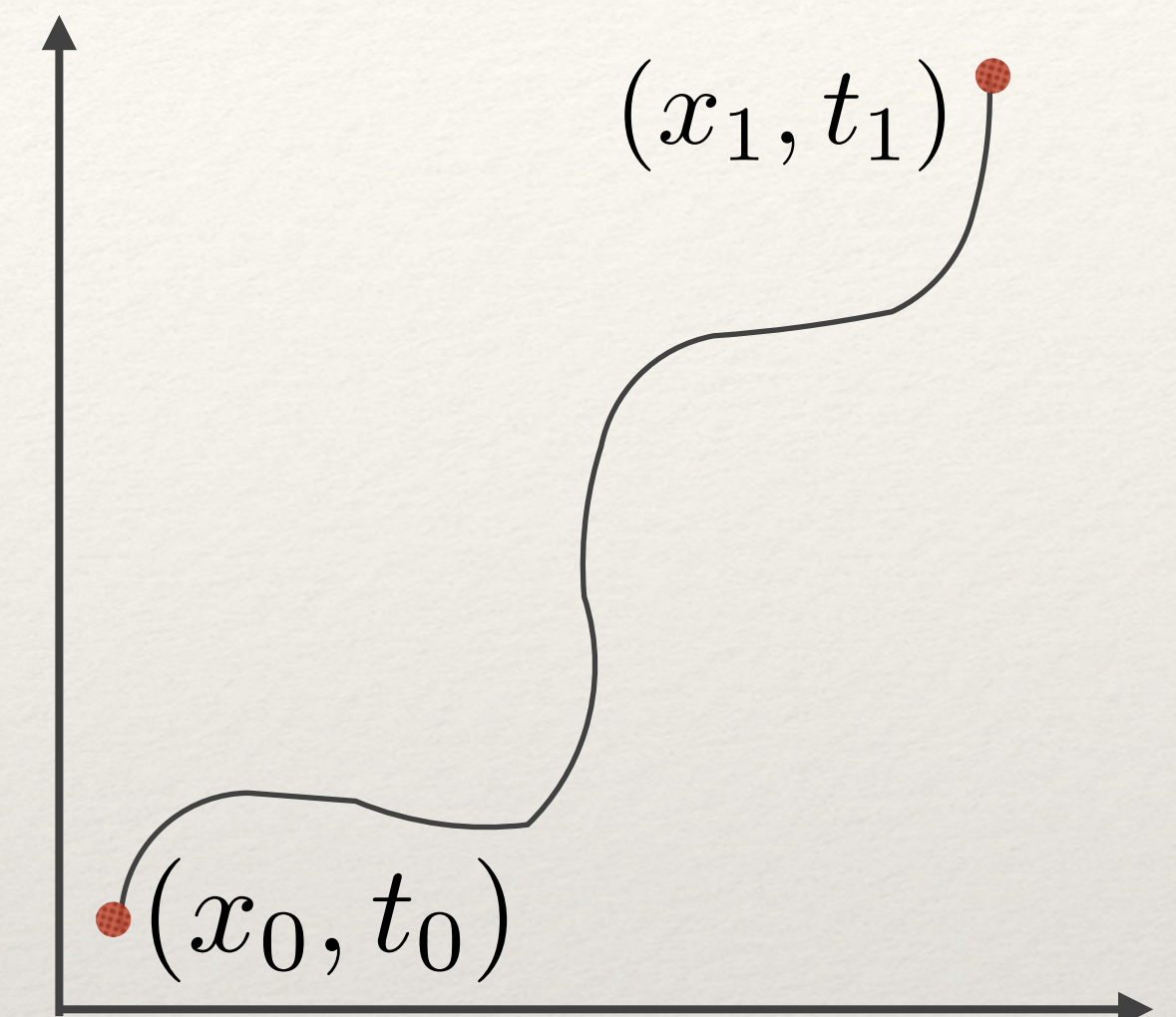
All the properties of particles can be measured with infinite precision.

The trajectory or dynamics of a particle can be measured via Newton's second law.

$$F = m\vec{a} = m \frac{d^2 \vec{r}}{dt^2}$$

The same equation of motion can be achieved via the principle of least action, or Lagrangian formalism.

$$\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) = 0.$$



Observation or experimentation on the particle hasn't been considered to have an effective consequence in classical physics

The departure from classical physics to special theory of relativity occurs for objects moving with very high velocities like the speed of light

Historical Interlude

The beginning of 1900 and early 1900 saw a big upheaval in the domain of science.

Special relativity was being discovered around the same time!

World War I (1914-1918) was a big event that shook the world.

The classical ideas started breaking which indicated the birth of a new subject. We will talk about some of those revolutionary experiments that occurred during the similar time.

They all led to the development of a new era of Quantum Physics (1900-1940)



Crisis in Physics I: Black Body Radiation (1900)

Spectrum of thermal radiation emitted by hot bodies depend on the composition of the body. However, one class of hot bodies emit thermal spectrum of universal characters. They are **black bodies**, not self luminous. Surface of them absorb all thermal radiation. Example, bismuth black.

Rayle-Jean's formula using classical theory predicts energy per unit volume of frequency

$$\rho_T(\nu) d\nu = \frac{8\pi\nu^2 kT}{c^3} d\nu$$

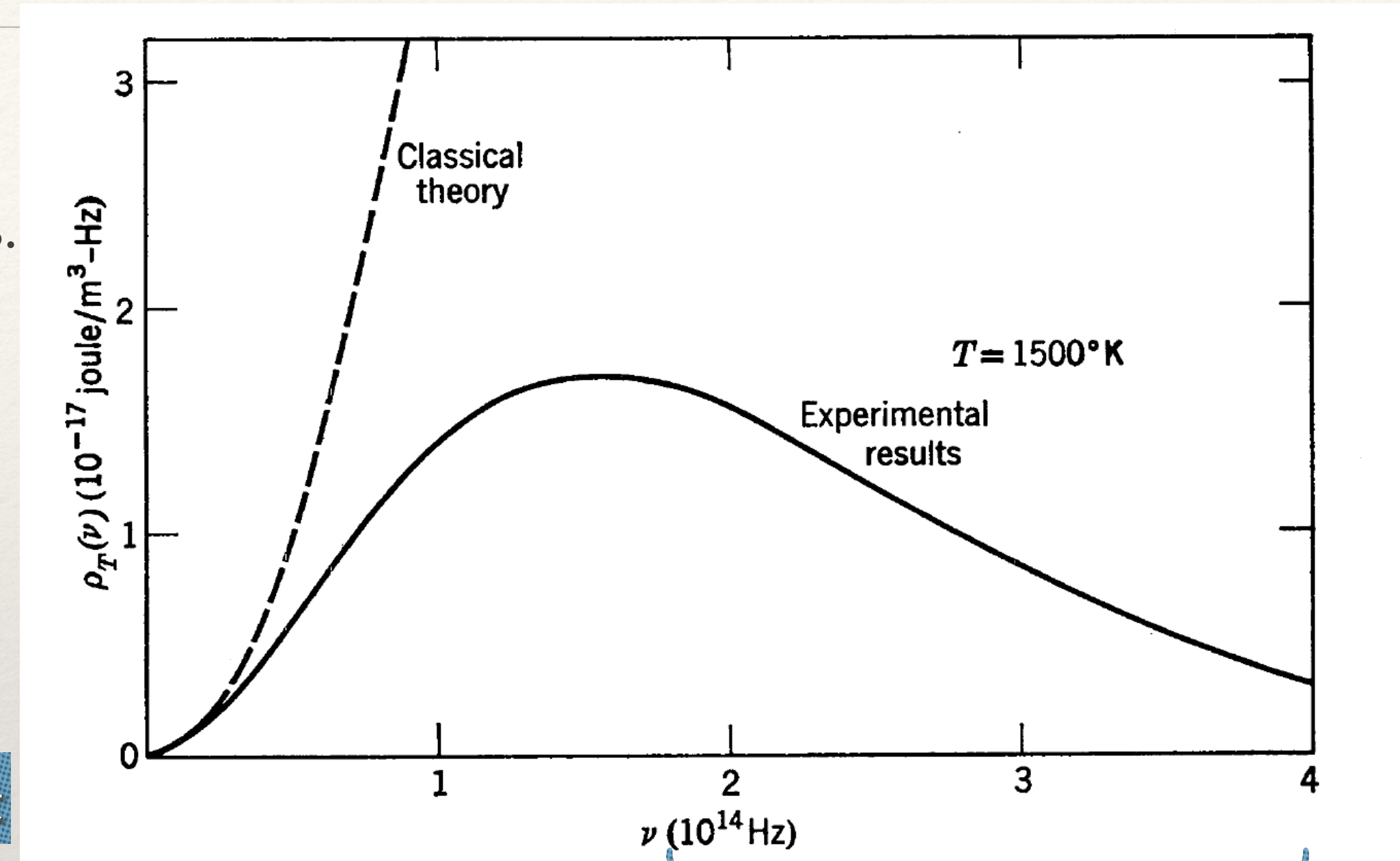
This can't explain the spectra!

$$k = 1.38 \times 10^{-23} \text{ joule/}^\circ\text{K}$$
 Boltzman's constant

Planck's theory:

$$\rho_T(\nu) d\nu = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{h\nu/kT} - 1} d\nu$$

This could successfully explain the blackbody radiation spectrum



No explanation: UV catastrophe

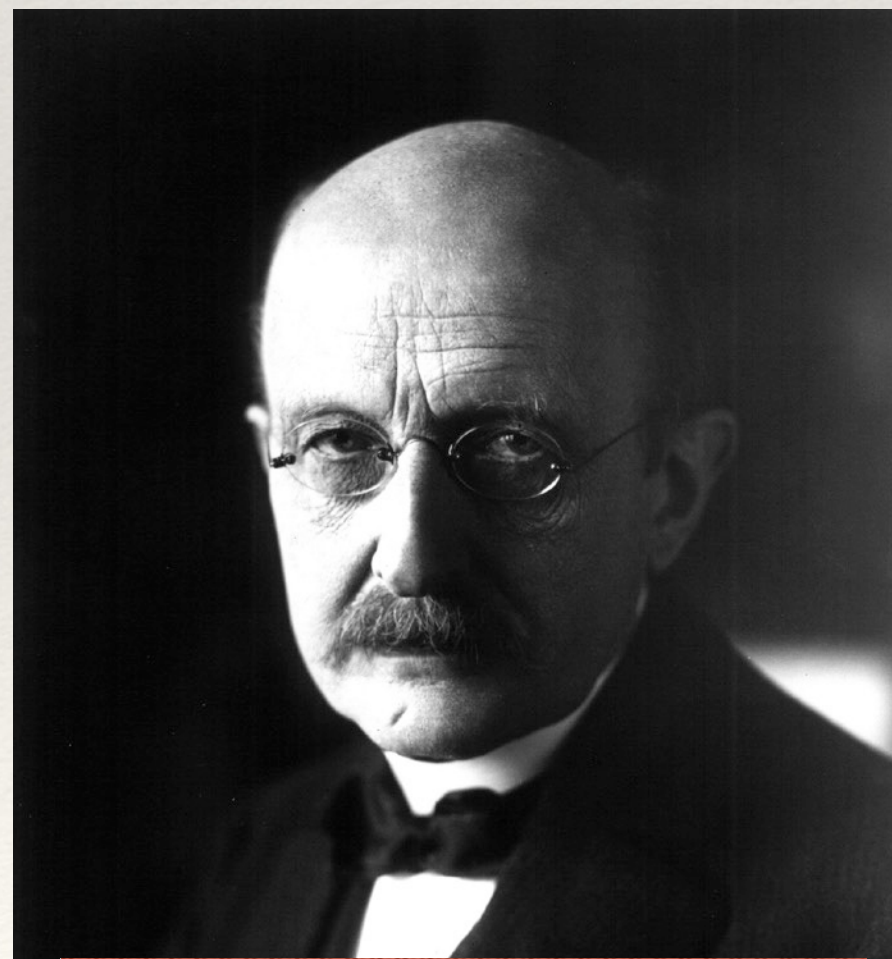
$$P(\mathcal{E}) = \frac{e^{-\mathcal{E}/kT}}{kT}$$

Probability density follows Boltzmann distribution

$$E = h\nu$$

$$h = 6.63 \times 10^{-34} \text{ joule-sec}$$
 Planck's constant

Radiation and absorption of a given frequency can only occur via 'quanta' of energy: 'an act of desperation'



Max Planck
Nobel prize: 1918

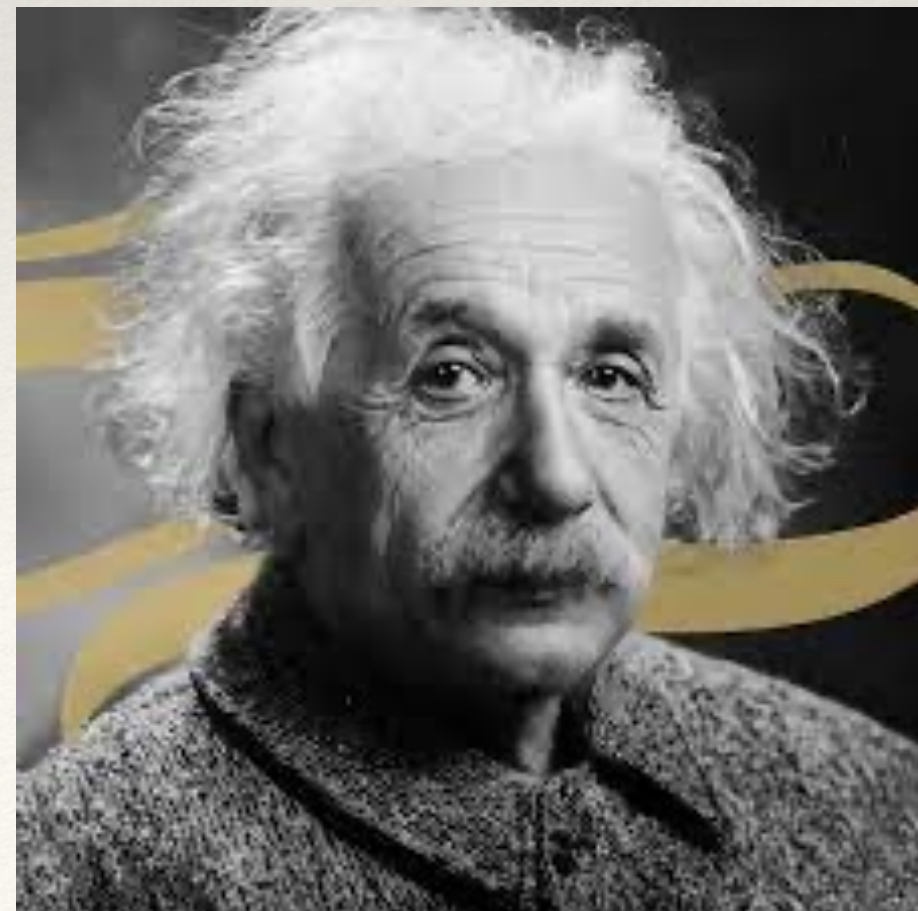
Crisis in Physics II: Photoelectric effect (1887-1905)

The photoelectric effect is the emission of electrons (photoelectrons) when electromagnetic radiation such as light hits a material. A photocurrent results thereafter.

First observed by Heinrich Hertz in 1887, but with some *unusual* properties.

- Only lights with **some desired frequency greater than a threshold frequency** can create photocurrent. Like blue lights with modest intensity can emit photoelectrons, but not the red light with very high intensity.
- The energy of photoelectrons doesn't depend on the intensity of the incident radiation, but depends on the frequency linearly.

They couldn't be explained via classical electromagnetic theory.



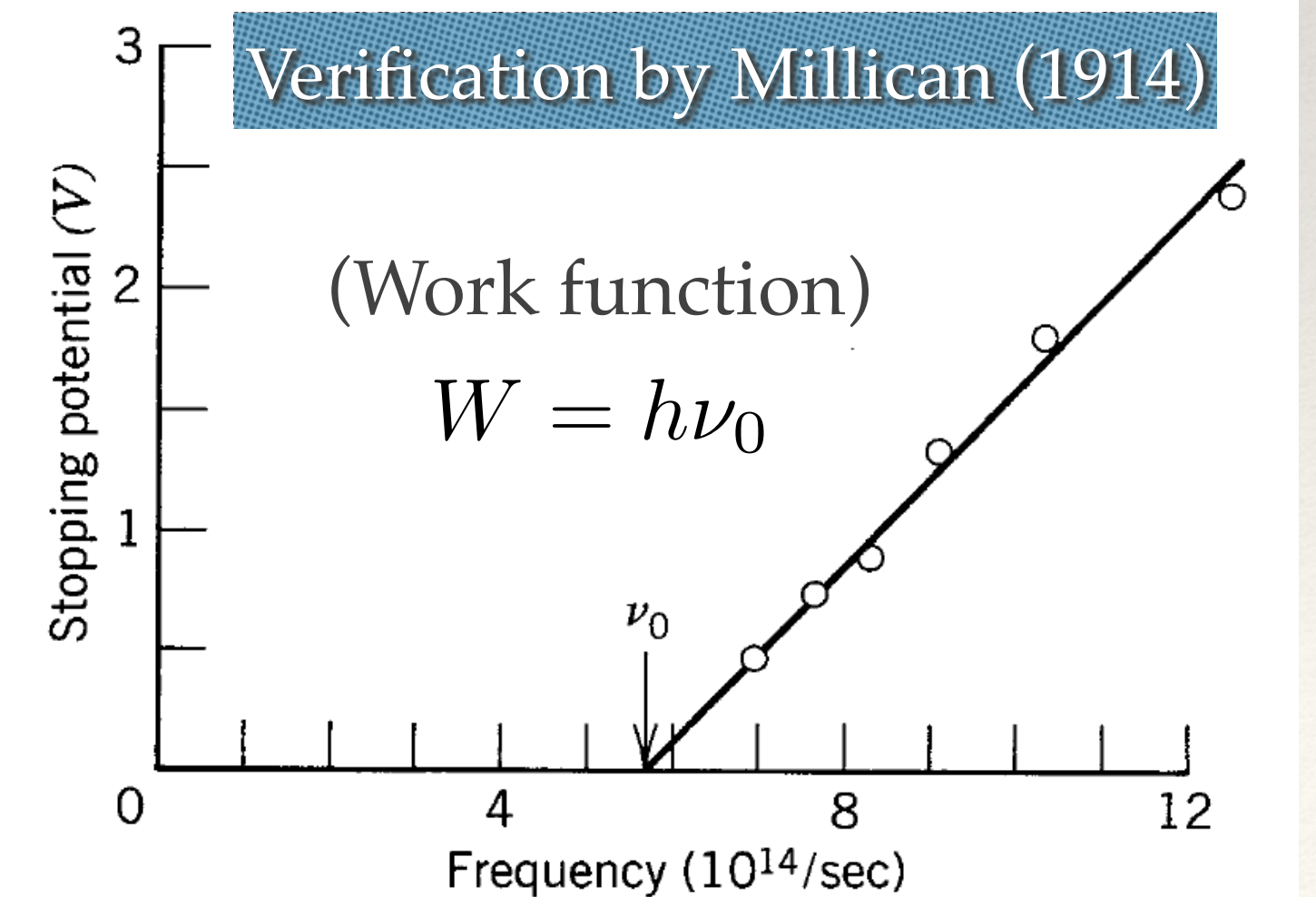
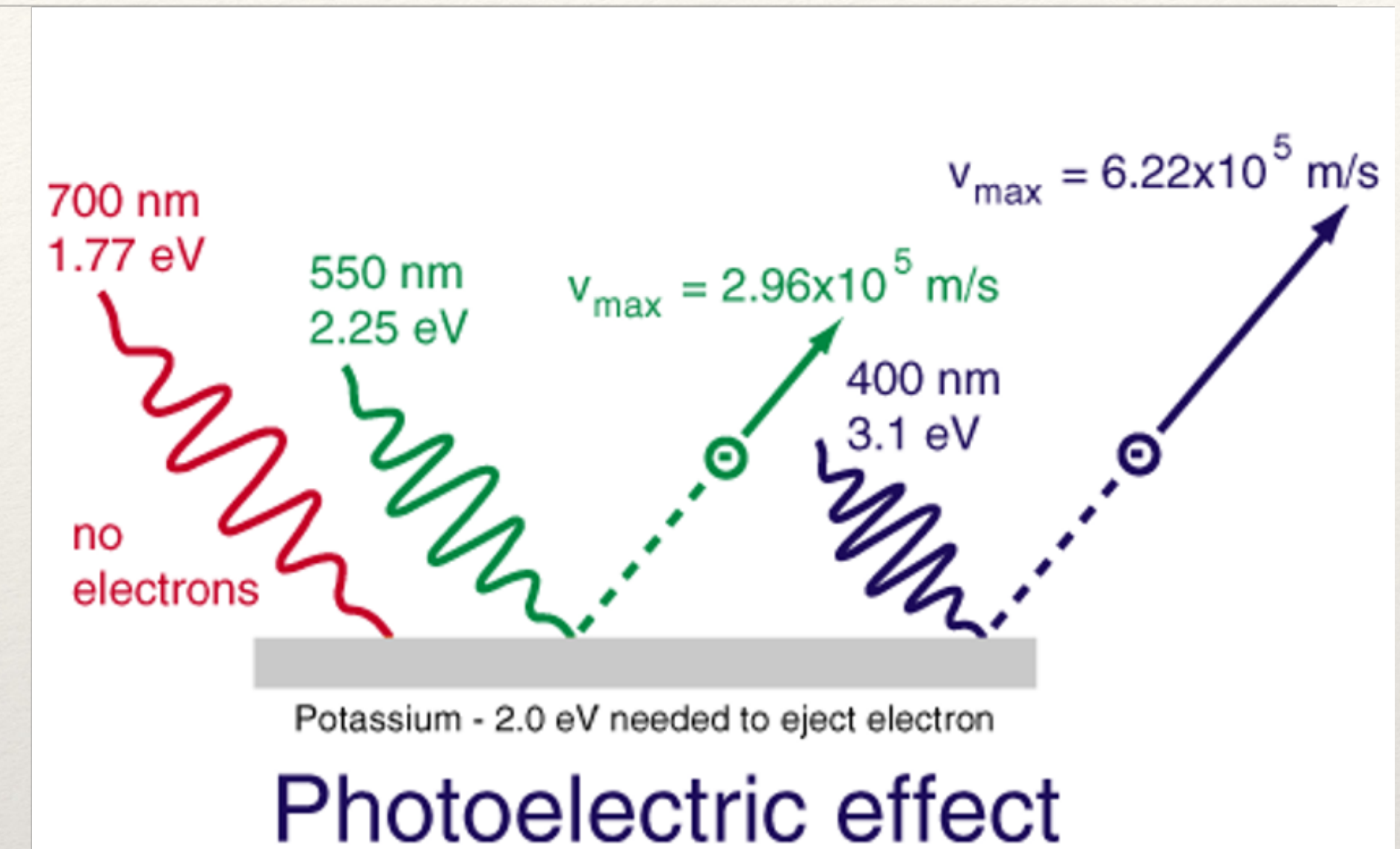
Einstein's proposal:

Nobel prize in 1921
(Not for relativity though)

$$\frac{1}{2}mv^2 = h\nu - W$$

$$E = h\nu$$

Beam of light is not wave, but swarm of discrete energy packets, Photons.



According to the assumption to be contemplated here, when a light ray is spreading from a point, the energy is not distributed continuously over ever-increasing spaces, but consists of a finite number of "energy quanta" that are localized in points in space, move without dividing, and can be absorbed or generated only as a whole.



George Bernard Shaw on Einstein

Crisis in Physics III: Atomic Stability



Niels Bohr
Nobel Prize: 1922

- Why don't atoms disintegrate in nanoseconds?
 - if electron is “orbiting”, it's accelerating (wiggling)
 - wiggling charges emit electromagnetic radiation (energy)
 - loss of energy would cause prompt decay of orbit

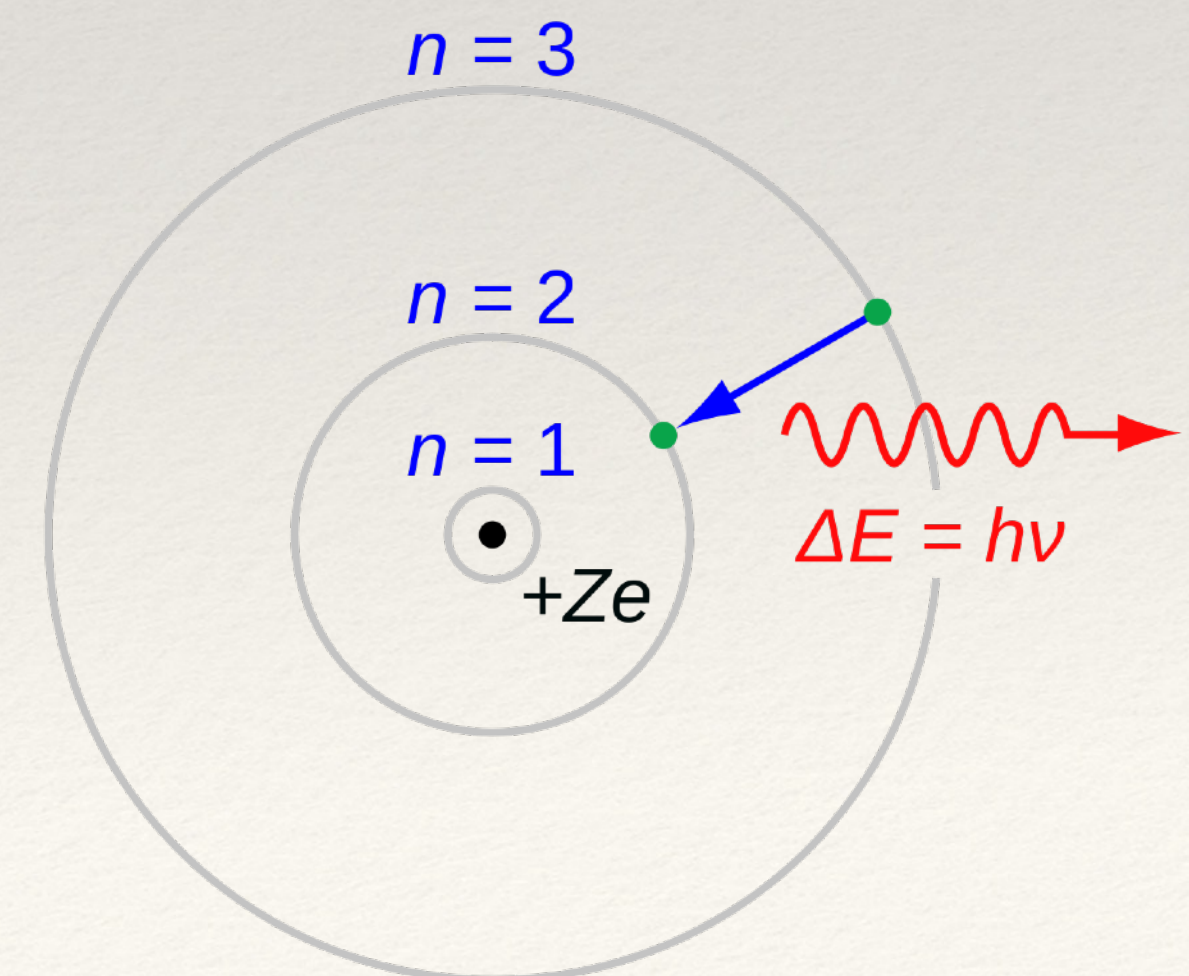
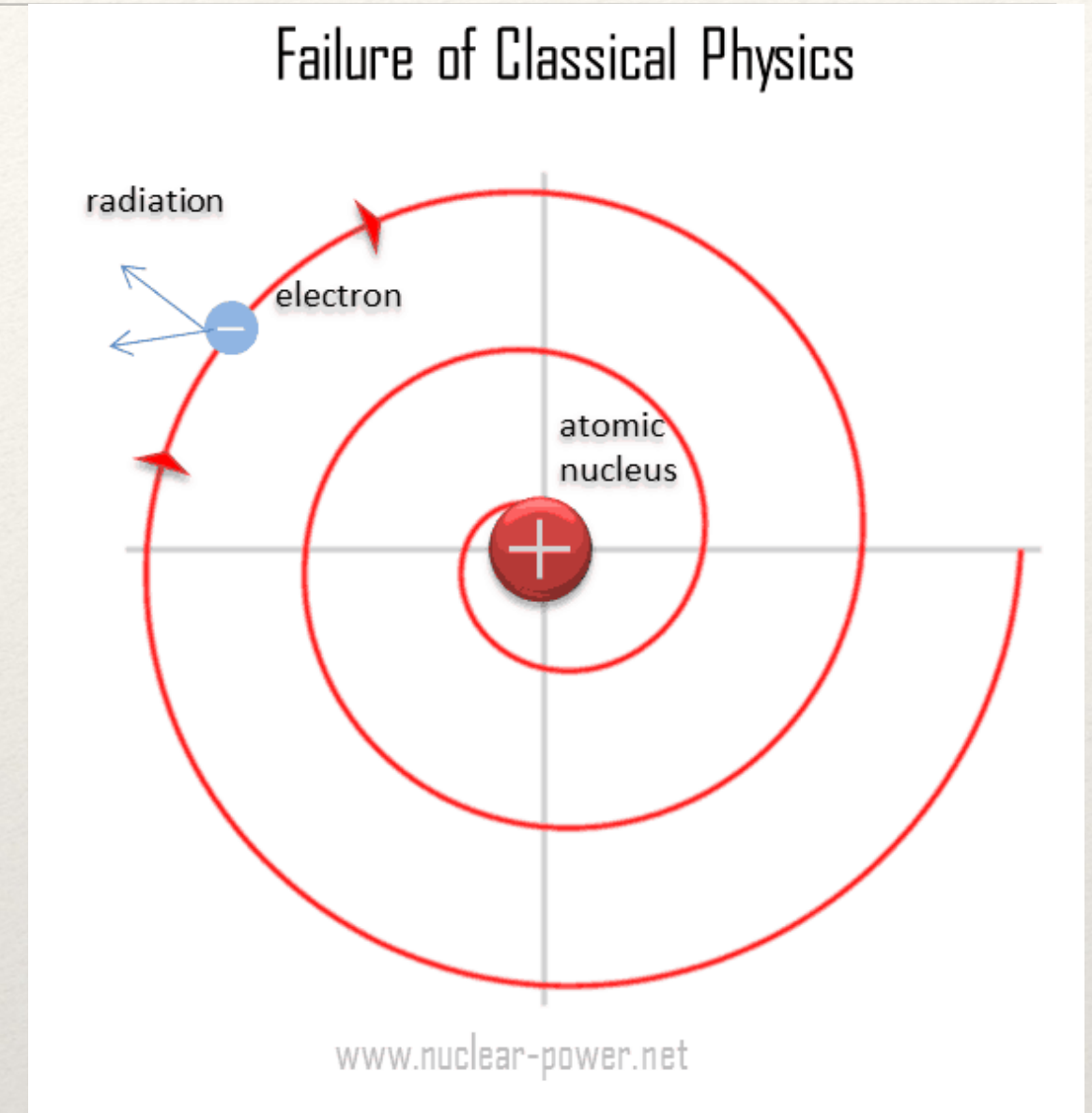
Bohr proposed a revolutionary model:
An atom with discrete (Quantum) states

Again, the idea was revolutionary and completely different from Classical physics.
But there was no concrete explanation.

An atomic system can only exist in a discrete set of stationary states, with discrete values of energy, and any change of the energy of the system, including emission and absorption of electromagnetic radiation must take place by a complete transition between two such stationary states.

$$h\nu = E_1 - E_2$$

$$mvr = n \frac{h}{2\pi}$$



Emission spectrum

- What caused spectra of atoms to contain discrete “lines”
 - it was apparent that only a small set of optical frequencies (wavelengths) could be emitted or absorbed by atoms
- Each atom has a distinct “fingerprint”
- Light only comes off at very specific wavelengths
 - or frequencies
 - or energies
- Note that hydrogen (bottom), with only one electron and one proton, emits several wavelengths

However, this formalism was also done in an ad-hoc way! No foundation to why this happens.

Resolution:
Rydberg's formula

$$\frac{1}{\lambda} = Ry \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Rydberg's
constant

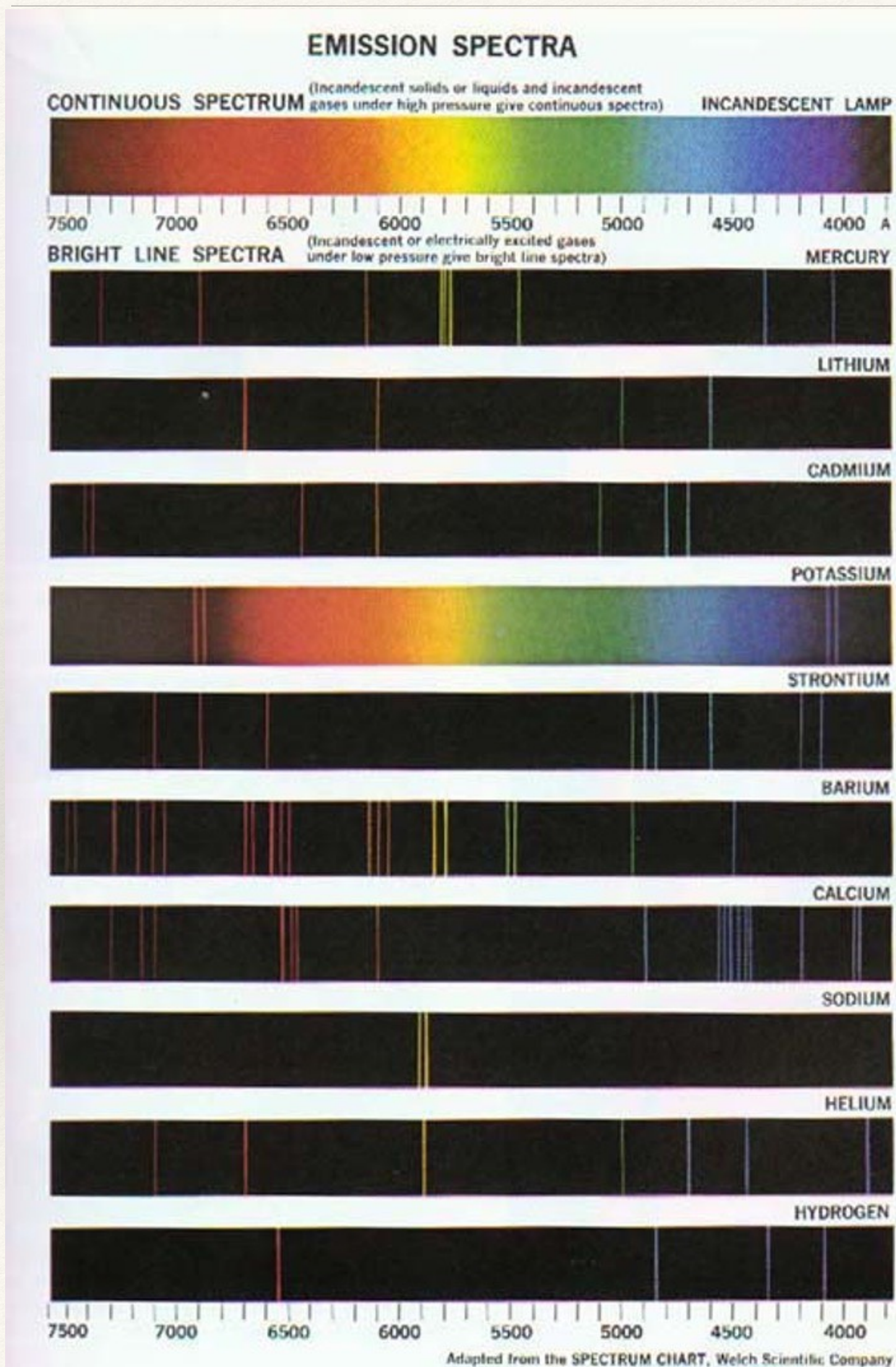
Upper
energy state

Lower
energy state

Bohr's model could explain the emission spectrum.

$$y = Z^2$$

Z: Atomic number



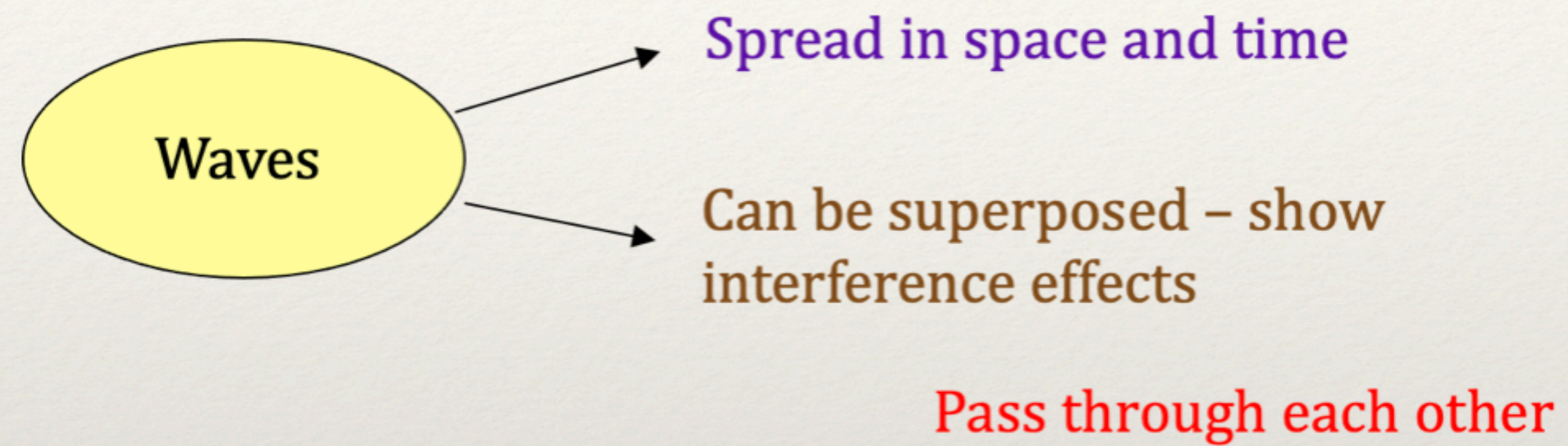
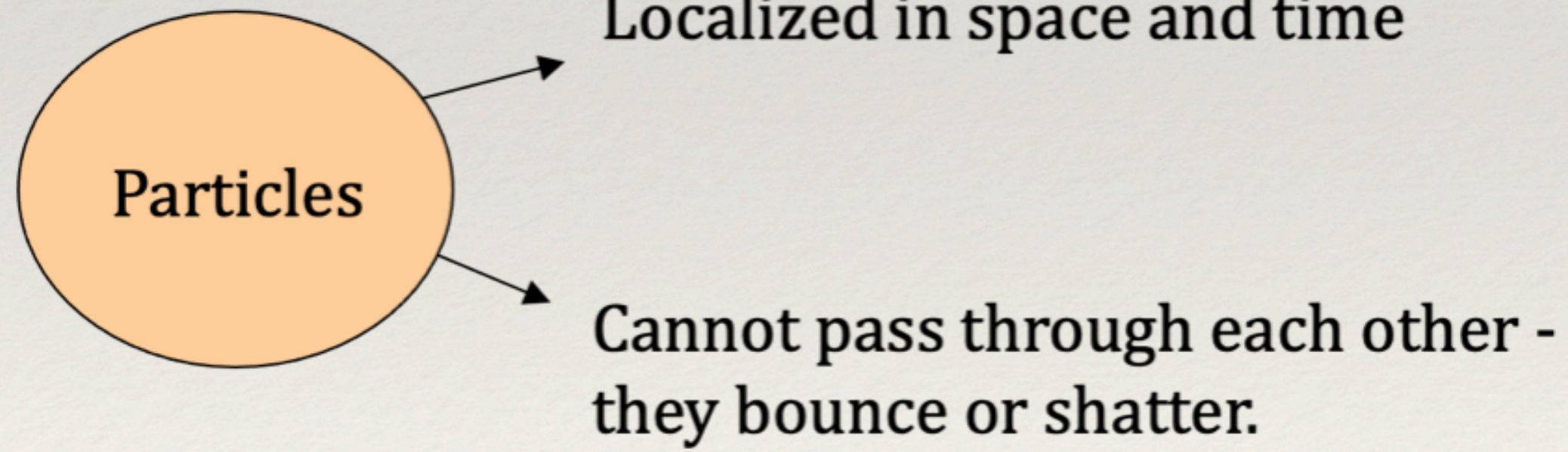
Dilemma

Particle like behaviour of light waves !

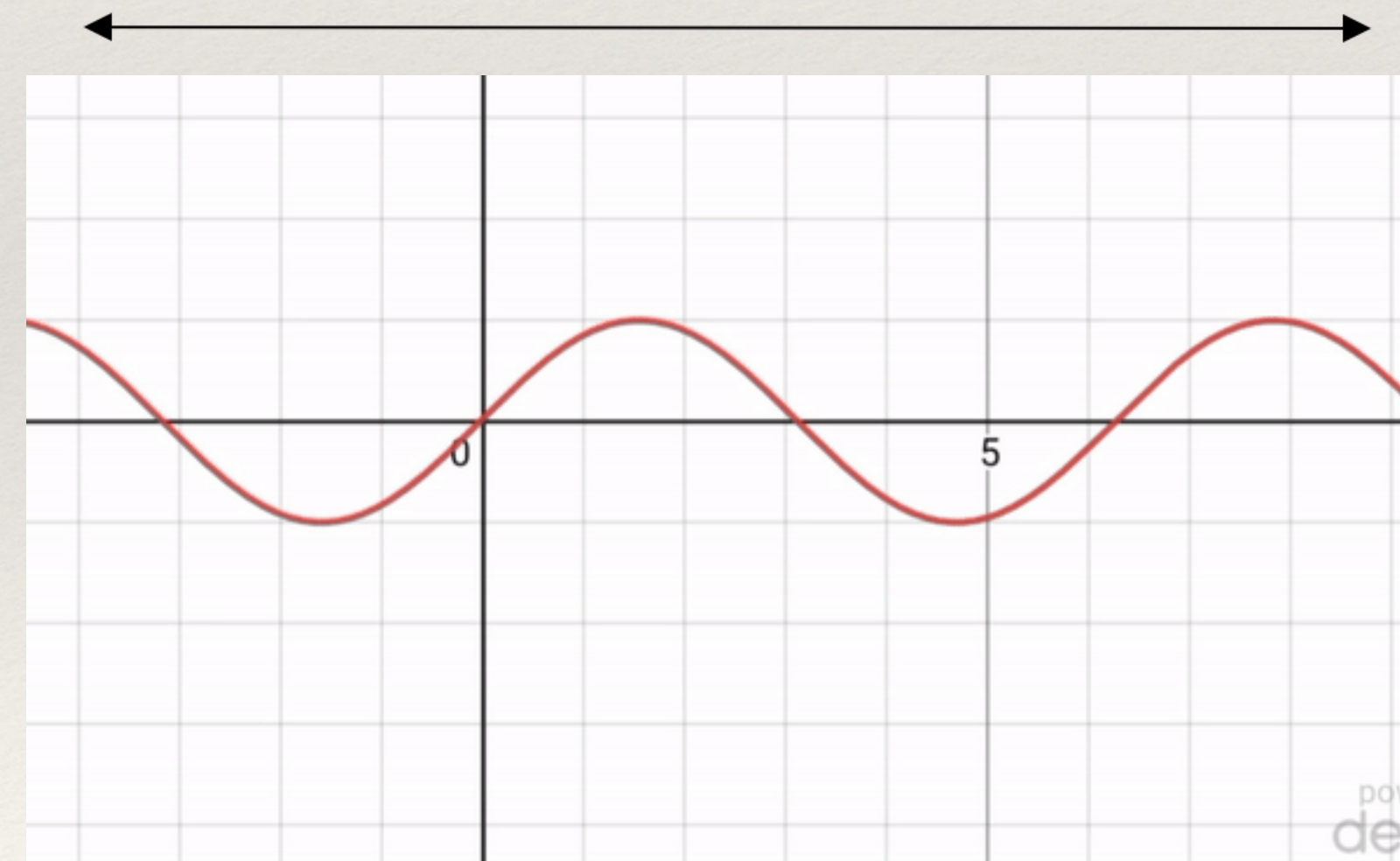
Is there a way to consistently describe the phenomena ?

Whether the particles can show wave nature ?

Waves and Particles



Falling Ball



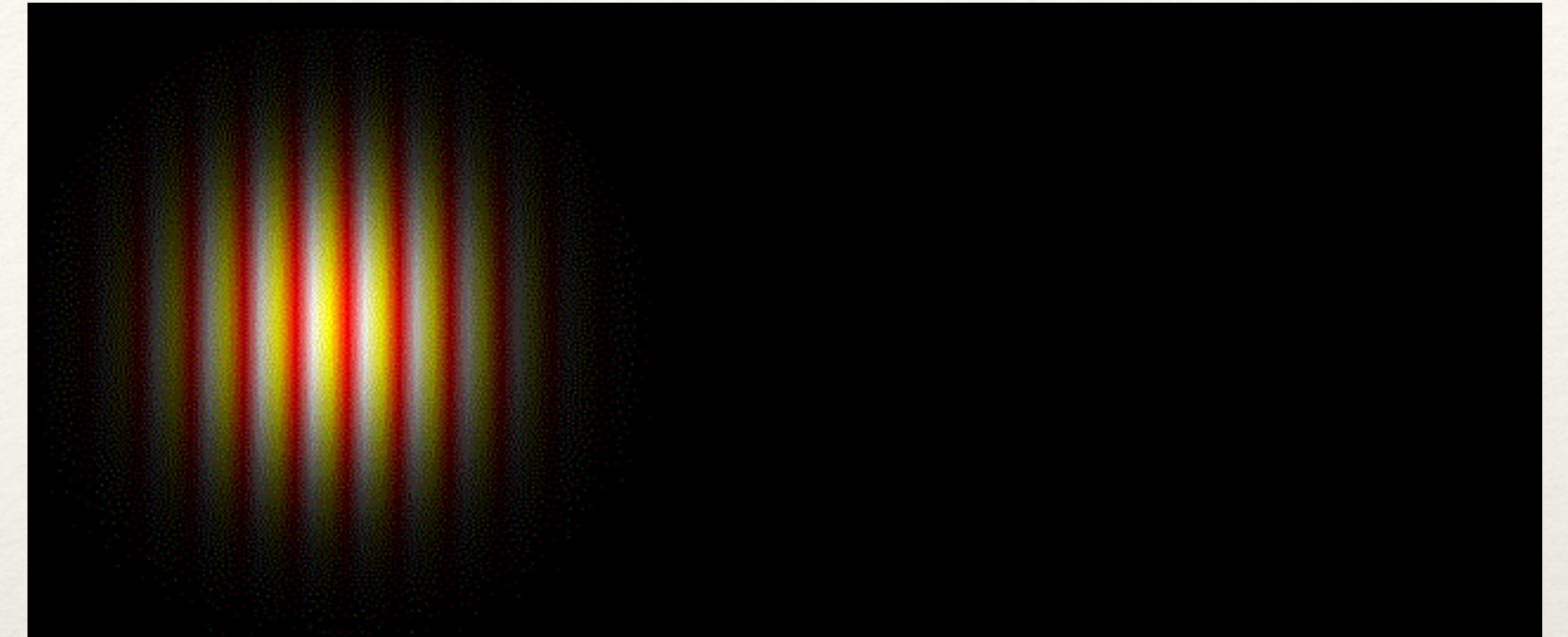
Wave particle duality



Louis de Broglie
Nobel prize: 1929

De Broglie's formulae was confirmed later via electron diffraction. We will talk about it later.

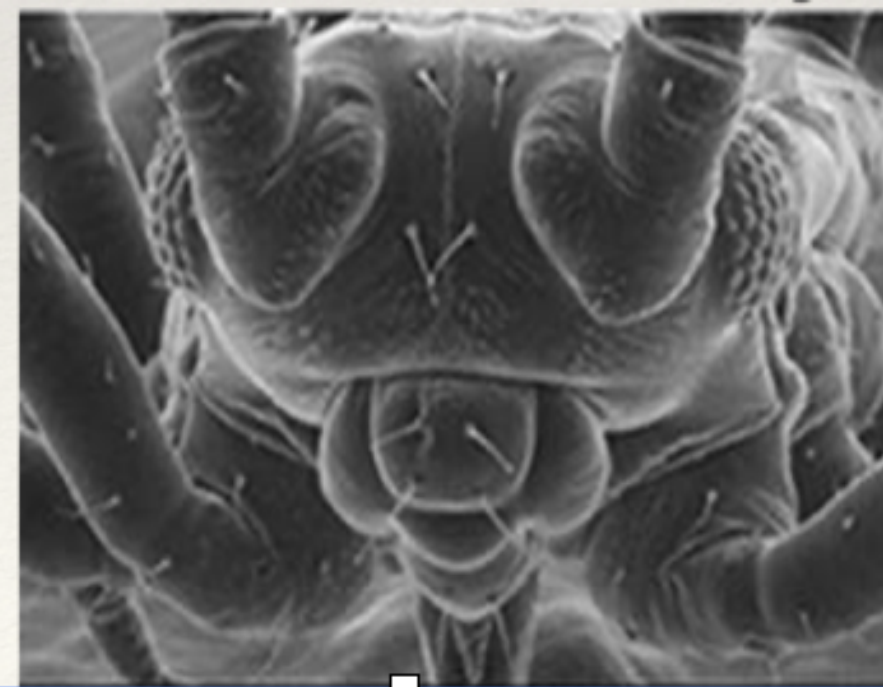
Louis de Broglie in 1923 proposed that matter particles should exhibit wave properties just as light waves exhibited particle properties. These waves have very small wavelengths in most situations so that their presence was difficult to observe



The quantum wavelength (de Broglie wavelength) of a particle is given by:

$$\lambda = \frac{h}{p}$$

- typical “quantum” objects:
 - electron (10^{-30} kg) at thermal velocity (10^5 m/s) $\rightarrow \lambda \approx 10^{-8}$ m
 - so λ is 100 times larger than an atom: very relevant to an electron!
- typical macroscopic objects
 - masses $\sim$ kg; velocities $\sim$ m/s $\rightarrow p \approx 1$ kg·m/s
 - $\lambda \approx 10^{-34}$ meters (too small to matter in macro environment!!)



Picture of a fly in electron microscope

$$E = h\nu = \frac{hc}{\lambda}$$

$$E = pc$$

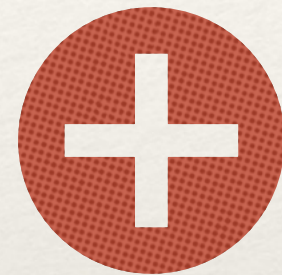
$$\lambda = \frac{h}{p}$$

A classic example: Compton scattering

One of the classic examples, where the wave-particle duality was verified was Compton scattering

Recall that we used for photon

$$E = h\nu = \frac{hc}{\lambda},$$

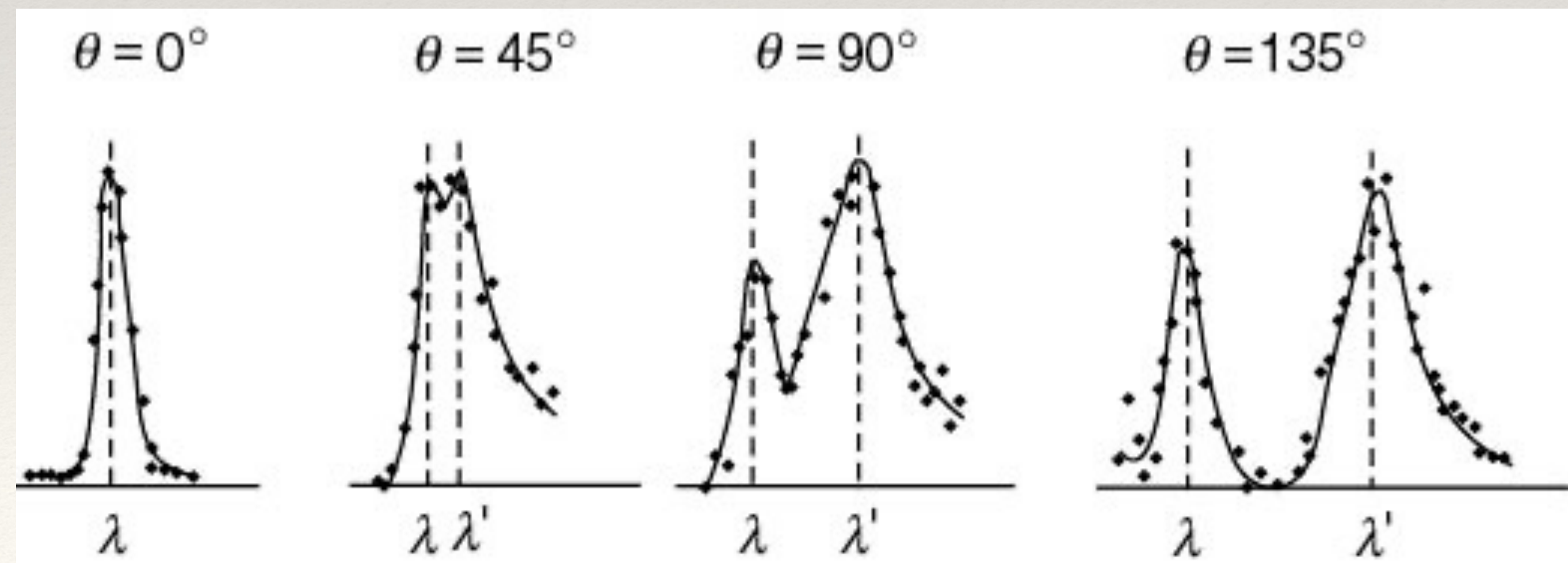
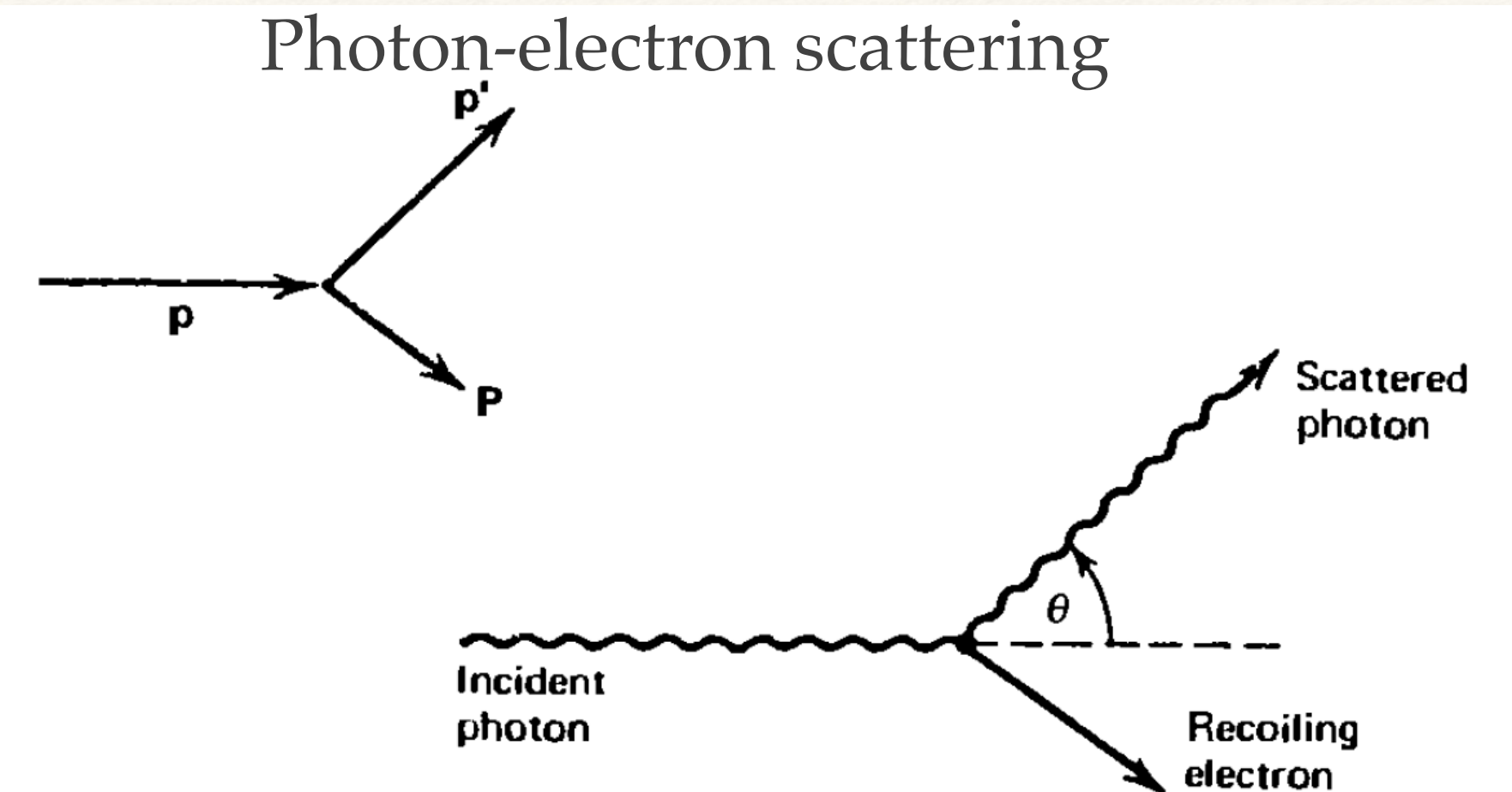


Relativistic energy-momentum conservation

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

The Compton shift of wavelength could be explained.

$$\Delta\lambda = \frac{h}{mc} (1 - \cos \theta)$$



Example

Q. Find the wavelength of the matter wave associated with a tennis ball moving at 110 km/h

A. First, $v \ll c$ so that, $p \approx m_0 v$ Take the mass of the tennis ball to be $m_0 \approx 58 \text{ g}$

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34} \text{ J.s}}{58 \times 10^{-3} \text{ kg} \times 110 \times 10^3 \text{ m}/(3600 \text{ s})}$$
$$= 3.73881 \times 10^{-34} \text{ m}$$

Compare this with the size of the hydrogen atom

$$= 53 \times 10^{-12} \text{ m}$$

The de Broglie wavelength of a tennis ball can be taken to be zero. This means a tennis ball has negligible wave-like properties.

Q. Find the wavelength of the matter wave associated with an electron moving at 110 km/h

First, $v \ll c$ so that, $p \approx m_0 v$

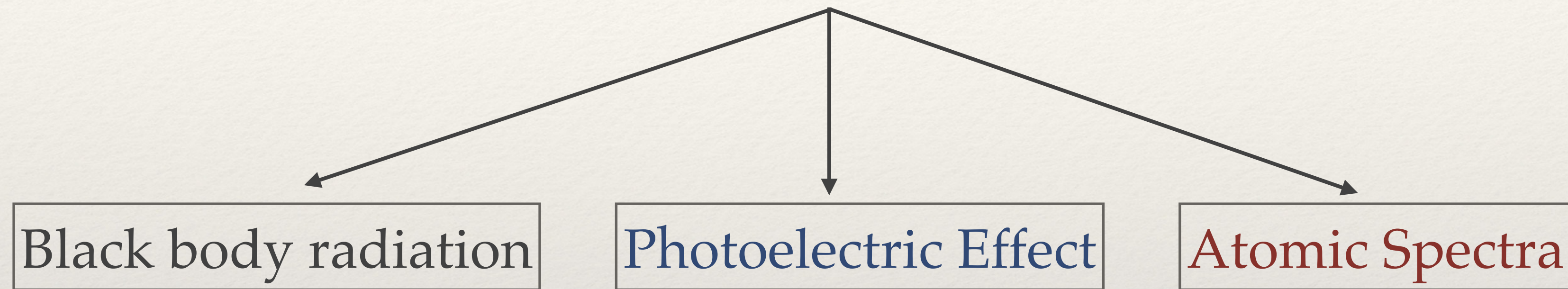
$$\lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{hc}{(m_0 c^2) \left(\frac{v}{c}\right)}$$

$$hc = 1.24 \text{ eV} \cdot \mu\text{m}$$
$$m_0 c^2 = 0.511 \text{ MeV}$$
$$\frac{v}{c} = 10^{-7}$$
$$\lambda = 24.3 \mu\text{m}$$

The typical size of a bacterium is $6 \mu\text{m}$.

Summary

We have talked about the experiments that established the uncanny connection between particles and waves.



Wave Particle Duality

$$\lambda = \frac{h}{p}$$